Project Development Phase

The source code lives in the repository root so the app keeps running with its current paths. This folder maps each module to its code.

Module to code map

Module	Backend	Frontend
Food Recognition	`backend/main.py`, `backend/nutrition.py`	`frontend/src/pages` (Food Recognition)
Freshness Detector	`backend/freshness.py`	`frontend/src/components/FreshnessDetector.jsx`
Hygiene Monitor	`backend/hygiene.py`	`frontend/src/pages` (Hygiene)
Smart Kitchen	`backend/smart_kitchen.py`, `backend/recipe.py`	`frontend/src/pages` (SmartKitchen)
Voice	`backend/voice.py`	Food Recognition page
Meal Planner	`backend/meal_ai.py`, `backend/meal_planner.py`	`frontend/src/pages` (MealPlanner)

Other folders

- `models/` trained weights and class list
- `notebooks/` experiments
- `scripts/` evaluation and checks
- `data/` datasets

Run

```
venv\Scripts\activate
uvicorn backend.main:app --reload --reload-dir backend
```

In a second terminal:

```
cd frontend
npm run dev
```

Model notes

- Image preprocessing: Resize 256, CenterCrop 224, ImageNet normalization.
- Own images for missing dishes: `own_train.zip`.
- Model files in `models/`: `foodverse_effb0.pt` (food recognition), `freshness_effb0.pt` and `freshness_binary.pt` (freshness). The `effb0` in the names indicates EfficientNet-B0.
- Freshness results on own photos: 77.8% (21 of 27). See `6.Project Testing/test_results.pdf`.